LIMITS OF A FUNCTION OF A REAL VARIABLE

Def'n

$\lim_{x \to a} f(x) = L$ if for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x - a| < \delta$, $|f(x) - L| < \varepsilon$.

Theorem

Let $f(x) = L$ be defined on an open interval containing $x = a$. Then, TFAE:

1) $\lim_{x \to a} f(x) = L$
2) If $\{x_n\}$ is a sequence where $\lim_{n \to \infty} x_n = a$ and $x \neq a$, then $\lim_{n \to \infty} f(x_n) = L$

Theorem

Uniqueness of Limits in Functions

If $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} f(x) = M$, then $L = M$

Theorem

Let f and g be functions, $a \in \mathbb{R}$. Assume $\lim_{x \to a} f(x) = L$, $\lim_{x \to a} g(x) = M$, $(L, M \in \mathbb{R})$.
Let $C \in \mathbb{R}$. Then,

1) If $f(x) = C$ for all $x \in \mathbb{R}$ then $L = C$.
2) $\lim_{x \to a} C f(x) = CL$.
3) $\lim_{x \to a} f(x) \pm g(x) = L \pm M$
4) $\lim_{x \to a} f(x) g(x) = LM$
5) $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{L}{M}$, if $M \neq 0$
6) $\lim_{x \to a} (f(x))^{\alpha} = L^{\alpha}$, for any $\alpha > 0$, $L > 0$
7) If $M = 0$ and $\lim_{x \to a} \frac{f(x)}{g(x)}$ exists then $L = 0$

Theorem

If $P(x)$ is a polynomial then $\lim_{x \to a} P(x) = P(a)$.

Theorem

Consider $\frac{P(x)}{q(x)}$ where P, q are polynomials

If $q(a) \neq 0$, then $\lim_{x \to a} \frac{P(x)}{q(x)} = \frac{P(a)}{q(a)}$

If $\lim_{x \to a} q(x) = 0$ and $\lim_{x \to a} P(x) \neq 0$, then

$\lim_{x \to a} \frac{P(x)}{q(x)}$ DNE.

If $\lim_{x \to a} q(x) = q(a) = 0 = P(a) = \lim_{x \to a} P(x)$,

then $(x - a)$ is a factor of both $P(x)$ and $q(x)$

Def'n

$\lim_{x \to a^+} f(x) = L$ if for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x - a| < \delta$ and $x > a$, then $|f(x) - L| < \varepsilon$.

$\lim_{x \to a^-} f(x) = L$ if for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x - a| < \delta$ and $x < a$, then $|f(x) - L| < \varepsilon$.

Theorem

Let $f(x)$ be defined on an open interval containing a, except possibly at $x = a$. Then, TFAE:

1) $\lim_{x \to a} f(x) = L$

2) $\lim_{x \to a^-} f(x) = L$ and $\lim_{x \to a^+} f(x) = L$

Theorem

Let f, q, h be functions defined on an open interval containing a except possibly at $x = a$. Then, if

$q(x) \leq f(x) \leq h(x)$ and $\lim_{x \to a} q(x) = L = \lim_{x \to a} h(x)$,

$\lim_{x \to a} f(x) = L$.

Theorem

Fundamental Trigonometric Limit

$\lim_{x \to 0} \frac{\sin(ax)}{ax} = 1$ for $a \in \mathbb{R}$

QUIZ #4, PRACTICE PROBLEMS #4

PRACTICE PROBLEMS

1. (a)

Let $\varepsilon > 0$ be given.
Let $\delta = \frac{\varepsilon}{2}$

Aside: $2\delta \le \varepsilon$
$\delta \le \frac{\varepsilon}{2}$

Then, if $0 < |x-3| < \delta$ we get

$$|(2x+1)-7| = |2x-6| = 2|x-3| < 2\delta \le \frac{2\varepsilon}{2} = \varepsilon$$

(b)

Let $\varepsilon > 0$ be given.
Let $\delta = \frac{\varepsilon}{9}$

Aside: $9\delta \le \varepsilon$
$\delta \le \frac{\varepsilon}{9}$

Then, if $0 < |x+1| < \delta$, we get

$$|(1-9x)-10| = |-9x-9| = 9|x+1| < 9\delta \le \frac{9\varepsilon}{9} = \varepsilon$$

(c)

Let $\varepsilon > 0$ be given.
Let $\delta > 3$

Then, if $0 < |x-3| < \delta$, we get

$$|3-3| = 0 < \delta \le \varepsilon$$

(d)

Let $\varepsilon > 0$ be given.
Let $\delta = \sqrt{\varepsilon}$

Aside: $\delta^2 \le \varepsilon$
$\delta \le \sqrt{\varepsilon}$

Then, if $0 < |x-2| < \delta$, we get

$$|x^2-4x+4| = |(x-2)^2| < \delta^2 \le (\sqrt{\varepsilon})^2 = \varepsilon$$

(e)

Let $\varepsilon > 0$ be given.
Let $\delta = \min\left\{1, \frac{36\varepsilon}{7}\right\}$

Aside: $\frac{7}{36}\delta \le \varepsilon$
$\delta \le \frac{36\varepsilon}{7}$

Then, if $0 < |x-3| < \delta$, we get

$$\left|\frac{1}{x^2} - \frac{1}{9}\right| = \left|\frac{9-x^2}{9x^2}\right| = \left|\frac{3+x}{9x^2}\right| |3-x| = \left|\frac{3+x}{9x^2}\right| |x-3|$$

Choose $\delta < 1$. Then, since $2 < x < 4$, we get

$$\left|\frac{3+x}{9x^2}\right| |x-3| < \left|\frac{7}{36}\right| |x-3| < \frac{7}{36}\delta \le \frac{7}{36} \cdot \frac{36\varepsilon}{7} = \varepsilon$$

2. Assume $L < 0$. Then, $\lim_{x \to a} f(x) = L$ if for $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x - a| < \delta$, $|f(x) - L| < \delta$

Choose $\varepsilon = \frac{|L|}{2}$. Hence, $|f(x) - L| < \frac{|L|}{2}$ and

$$L - \frac{|L|}{2} < f(x) < L + \frac{|L|}{2}$$

Since $L < 0$, we get $f(x) < L + \frac{|L|}{2} < 0$.

However, this proves a contradiction to our initial assumption that $f(x) > 0$. Thus, $L \geq 0$.

3. (a) Let $x_n = 3 + \frac{1}{n}$. Clearly $x_n \to 3$ as $n \to 3$. Then,

$$f(x) = \frac{1}{x-3} = \frac{1}{(3 + \frac{1}{n}) - 3} = \frac{1}{\frac{1}{n}} = n.$$

Since $\lim_{n \to \infty} f(x_n) = \infty$, $\lim_{x \to 3} \frac{1}{x-3}$ DNE.

(b) Let $x_n = \frac{1}{n}$. Clearly $x_n \to 0$ as $n \to \infty$. Then,

$$f(x) = \ln|x| = \ln\left|\frac{1}{n}\right| = \ln(1) - \ln|n| = -\ln|n|$$

Since $\lim_{n \to \infty} f(x_n) = -\infty$, $\lim_{x \to 0} \ln|x|$ DNE.

4. Let $x_n = 3 + \frac{1}{n}$, $y_n = 3 - \frac{1}{n}$. Clearly $x_n \to 3$, $y_n \to 3$, $x_n \neq 3$, $y_n \neq 3$ for all $n \in \mathbb{N}$. Then,

$$f(x_n) = \frac{(3 + \frac{1}{n})^2 - 9}{|(3 + \frac{1}{n}) - 3|} = \frac{9 + \frac{6}{n} + \frac{1}{n^2} - 9}{|\frac{1}{n}|} = |n|\left(\frac{6}{n} + \frac{1}{n^2}\right) = 6 + \frac{1}{n}$$

Thus, $\lim_{n \to \infty} 6 + \frac{1}{n} = 6 + 0 = 6$. Similarly,

$$f(y_n) = \frac{(3 - \frac{1}{n})^2 - 9}{|(3 - \frac{1}{n}) - 3|} = \frac{9 - \frac{6}{n} + \frac{1}{n^2} - 9}{|-\frac{1}{n}|} = |-n|\left(\frac{-6}{n} + \frac{1}{n^2}\right) = -6 + \frac{1}{n}$$

Thus, $\lim_{n \to \infty} -6 + \frac{1}{n} = -6 + 0 = -6$.

Since $\lim_{n \to \infty} f(x_n) \neq \lim_{n \to \infty} f(y_n)$, $\lim_{x \to 3} \frac{x^2 - 9}{|x - 3|}$ DNE.

5. (a)

$$= \lim_{x \to 4} \frac{(x+4)\cancel{(x-4)}}{\cancel{x-4}} = \lim_{x \to 4} x + 4 = 4 + 4 = 8$$

(b)

$$= \lim_{x \to 2} \frac{(x-2)^3}{x-2} = \lim_{x \to 2} (x-2)^2 = 0^2 = 0$$

(c)

$$= \lim_{x \to 16} \frac{\sqrt{x} - 4}{x - 16} \left(\frac{\sqrt{x} + 4}{\sqrt{x} + 4} \right)$$

$$= \lim_{x \to 16} \frac{\cancel{x-16}}{(\cancel{x-16})(\sqrt{x} + 4)}$$

$$= \lim_{x \to 16} \frac{1}{\sqrt{x} + 4}$$

$$= \frac{1}{8}$$

(d)

$$= \lim_{x \to \frac{\pi}{2}} \frac{\sec^2(x)}{\sec^2(x)} = 1$$

(e)

Since $\lim_{x \to 3^+} \frac{3x - 9}{3 - x} = \lim_{x \to 3^+} \frac{-3(3-x)}{3-x} = -3$

and $\lim_{x \to 3^-} \frac{-3x + 9}{3 - x} = \lim_{x \to 3^-} \frac{3(3-x)}{3-x} = 3$,

$\lim_{x \to 3} \frac{|3x - 9|}{3 - x}$ DNE.

(f)

Since $\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} (x-1) - 1 = -1$ and

$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} -(x-1) - 1 = -1$,

$\lim_{x \to 1} f(x) = -1$.

6.

$\lim_{x \to b^-} f(x) = \lim_{x \to b^-} 1 + \frac{1}{x} = 1 + \frac{1}{b}$ and

$\lim_{x \to b^+} f(x) = \lim_{x \to b^+} 1 + x = 1 + b$

So, if $\lim_{x \to b} f(x)$ exists, $1 + \frac{1}{b} = 1 + b$. Then,

$$1 + \frac{1}{b} = 1 + b \Rightarrow \frac{1}{b} = b \Rightarrow 1 = b^2 \Rightarrow b = \pm 1$$

Thus, $L = 2$ or $L = 0$. Since the one-sided limits do not agree for any other value of b, $\lim_{x \to b} f(x)$ DNE.

7. (a)

$$= \lim_{x \to 0} \frac{\sin(x)}{x} \cdot \frac{\sin(x)}{x} = 1 \cdot 1 = 1$$

(b)

$$= \lim_{x \to 0} \left(\frac{1 - \cos(x)}{x^2}\right)\left(\frac{1 + \cos(x)}{1 + \cos(x)}\right)$$

$$= \lim_{x \to 0} \frac{1 - \cos^2(x)}{x^2(1 + \cos(x)}$$

$$= \lim_{x \to 0} \frac{\sin^2(x)}{x^2(1 + \cos(x))}$$

$$= \lim_{x \to 0} \frac{\sin^2(x)}{x^2} \cdot \lim_{x \to 0} \frac{1}{1 + \cos(x)}$$

$$= 1 \cdot \frac{1}{2}$$

$$= \frac{1}{2}$$

(c)

$$= \lim_{x \to 2} \frac{\sin(3(x^2 - 4))}{x - 2} \left(\frac{3(x + 2)}{3(x + 2)}\right)$$

$$= \lim_{x \to 2} \frac{\sin(3(x^2 - 4))}{(3(x^2 - 4))} (3(x + 2))$$

$$= 1 \cdot (3(2 + 2))$$

$$= 12$$

(e)

$$= \lim_{x \to 0} \frac{\sin(x^2)}{\sqrt{|x^3|}} \left(\frac{\sqrt{|x|}}{\sqrt{|x|}}\right)$$

$$= \lim_{x \to 0} \frac{\sin(x^2)}{\sqrt{|x^4|}} (\sqrt{|x|})$$

$$= \lim_{x \to 0} \frac{\sin(x^2)}{x^2} (\sqrt{|x|})$$

$$= 1 \cdot \sqrt{|0|}$$

$$= 0$$

8. (a) F
(b) T
(c) F
(d) T
(e) F

9 (a) T
(b) T
(c) F
(d) F
(e) F